

NIRIKSHAK AI — Tech Stack Report

Comprehensive Technology & Architecture Specification

Executive Overview

NIRIKSHAK AI is an advanced AI/ML surveillance and monitoring platform designed for the **MPLADS** (Member of Parliament Local Area Development Scheme) ecosystem. The architecture fuses five distinct analytical layers: *Deterministic Rules*, *Statistical Analytics*, *Unsupervised Machine Learning*, *Natural Language Processing*, and *Predictive AI* into a unified risk score engine with real-time alerting and intuitive interactive dashboards.

Category	Technologies / Libraries	Role & Purpose
Frontend Framework	Next.js 16 (App Router) React 19, React DOM 19	Modern SSR/SSG full-stack UI architecture for interactive web dashboards.
Frontend Language & Tools	TypeScript 5 ESLint v9	Type safety, developer productivity, and strict code quality enforcement.
Styling & UI Components	Tailwind CSS v4 PostCSS Lucide React	Utility-first styling, responsive layout math, and accessible UI icon sets.
Data Visualization	Recharts v3	Interactive financial analytics, geographical trends, and risk score charts.
Backend API Engine	FastAPI Uvicorn ASGI Pydantic & Pydantic-Settings	High-performance asynchronous Python REST API framework with automated OpenAPI docs.
Database & ORM	SQLite (Prototype) PostgreSQL (Production) SQLAlchemy 2.0 Alembic	Relational persistence, database schema migrations, and structured query management.
Unsupervised ML & Anomaly	Scikit-Learn (IsolationForest, LOF)	Multivariate anomaly detection for expenditure spikes and vendor risk behaviors.
NLP & Duplicate Detection	Sentence-Transformers HuggingFace ANN Search	Text embedding generation and semantic similarity matching for duplicate works.
Entity Resolution	Python difflib (SequenceMatcher) Custom Regex Engine	MP name normalization, vendor alias resolution, and work ID parsing.
Statistical Engines	EWMA, Rolling Z-Score Mann-Kendall Trend PELT Change-Point HHI & Gini Index	Time-series spend trend analysis, velocity tracking, and concentration risk calculation.

Category	Technologies / Libraries	Role & Purpose
Data Processing	Pandas, NumPy PyYAML Great Expectations	Data cleaning, header normalization, config management, and automated schema testing.
Network & Graph Analytics	NetworkX	Graph community detection and centrality scoring for MP-Vendor-IDA networks.
Async Tasks & Tracking	Celery Redis Broker MLflow	Background task orchestration, event broker, and machine learning experiment tracking.

Key Architectural Highlights

- **5-Layer Risk Fusion:** Fuses Rules, Statistics, Unsupervised ML, NLP Embeddings, and Survival Analysis into a single calibrated risk score (0-100).
- **Canonical Feature Store:** Standardized feature tables (work, vendor, MP, transaction) decouple raw ingestion from ML model evaluation.
- **Zero Hardcoded Thresholds:** Every risk parameter and threshold is configurable via YAML or dynamic learned baselines.
- **Explainable AI (XAI):** Every risk score is accompanied by human-readable evidence strings and detailed factor breakdowns.